

What is the difference between "augmented matrix" and "matrix equation" of a linear system? When do we use which?

Answer:

The "augmented matrix" and "matrix equation" of a linear system provide different ways to represent the same set of linear equations, each useful for specific purposes:

Matrix Equation Form

- **Structure:** This is expressed as $\mathbf{Ax} = \mathbf{b}$ where:
 - $\mathbf{A}$ is the matrix of coefficients.
 - $\mathbf{x}$ is the vector of variables.
 - $\mathbf{b}$ is the vector of constants.
- **Use:** It's helpful for theoretical discussions, applying linear algebra concepts like invertibility, using eigenvalues, determinants, and transformations.

Augmented Matrix Form

- **Structure:** This combines the matrix of coefficients and the constants vector into one matrix, written as $(\mathbf{A} \mid \mathbf{b})$
- **Use:** It is practical for performing row operations like Gaussian or Gauss–Jordan elimination for solving the linear system. It's easier to manage and visualize these operations without explicitly involving variables.

When to Use Which

- **Matrix Equation:** Best for theoretical applications and compact representations, especially when you wish to apply matrix operations or discuss properties like invertibility.
- **Augmented Matrix:** Best for manual or computational solving of the system using row operations to reach row echelon form or reduced row echelon form.

These two forms are complementary, often used together in problem solving: starting with the matrix equation to set up the problem, transitioning to the augmented matrix for solution, and interpreting results back in the context of the matrix equation.